

KWAKU BOATENG

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EDUCATION

Grambling State University, *Bachelor of Science in Computer Science and Cloud Computing* - GPA 3.92/4.0

December 2027

Relevant Coursework: Data Structures & Algorithms, Database Systems, AI, Statistical Analysis, Computer Networks, Systems Programming

Certificates: IBM Artificial Intelligence Practitioner, CompTIA Security+, CodePath TIP 102

Honors: Presidential Scholar, Top 100 Governor's Scholarship Recipient, Earl Lester's Honors Scholar

SKILLS

- **Programming Languages:** Java, Python, Go, C++, JavaScript/TypeScript, OCaml, SQL
- **Technologies & Tools:** AWS, Docker, Kafka, CUDA, Spring Boot, React, Node.js, Flask, Django, Git, Splunk

WORK EXPERIENCE

All State Insurance, *Software Engineering Intern (Product Focus)*

05/2026 – 08/2026 | Northbrook, Illinois

- Engineer full-stack web applications using Java and Spring Boot, increasing feature deployment efficiency by 15% across 3 core pipelines.
- Diagnose backend bottlenecks using root cause analysis, writing automated Java/JUnit tests that cut manual debugging time by 25%.
- Participate in Agile ceremonies to architect reliable backend components, optimizing SQL database query speeds by 30%.

Jane Street, *Software Engineering Track (IN FOCUS)*

01/2026 – 01/2026 | New York City, NY

- Learned OCaml and core concepts of electronic trading systems using Jane Street's libraries and tools during an intensive 1-week track.
- Implemented a functional trading program to execute automated strategies against a simulated exchange of 50+ concurrent participants.
- Built 2 distinct backend and UI components for a computer game to master functional programming design patterns.

Codveda Technologies, *Software Engineering Intern (Backend Focus)*

07/2025 – 08/2025 | remote, USA

- Engineered CI/CD pipeline extensions using Python to streamline code validation, cutting production deployment runtime by 35%.
- Designed and integrated internal RESTful APIs to synchronize cross-team repository states, increasing framework data reliability.
- Optimized backend data caching layers and query indexing to handle high-throughput system performance metrics with minimal latency.

PROJECTS

RaftMage – Fault-Tolerant Distributed Key-Value Store | (Go, Raft Consensus, gRPC, Protobuf)

- Architected a distributed, sharded key-value store in Go utilizing the Raft consensus protocol to ensure strong data consistency.
- Implemented a custom gRPC API with Protobuf to facilitate low-latency inter-node communication and state machine replication.
- Designed fault-tolerant recovery mechanisms to handle network partitions and node crashes with zero data loss.

TechDigest – Full-Stack Engineering | (React, Tailwind CSS, Flask, OpenAI API, Netlify, GitHub Actions)

- Architected a responsive full-stack web application using React and Flask to automate data aggregation and summarization for 500+ users.
- Integrated asynchronous processing to handle heavy OpenAI API payloads, optimizing client-side rendering speeds and UI responsiveness.
- Established a CI/CD pipeline via GitHub Actions for automated testing, deploying to production with 99.9% uptime validation.

Safelink - Decentralized Mesh Messaging System | (BLE/WiFi Direct Mesh, Web API, Leaflet, AES-256, Vite)

- Contributed to building an offline-first peer-to-peer mesh messaging system for zero-infrastructure environments
- Implemented secure messaging features using AES-256 encryption with HMAC verification to ensure data integrity
- Assisted in designing routing logic and retry mechanisms to improve message delivery in unstable networks

DailySync – Real-Time Streaming & Telemetry Pipeline | (Kafka, SQL, Pandas, Grafana)

- Architected real-time data ingestion pipelines using Kafka to stream performance metrics for 10K+ concurrent users.
- Designed fault-tolerant message queues and automated retry logic to eliminate data loss and guarantee high availability.
- Integrated Grafana dashboards with optimized SQL query engines to visualize streaming telemetry with sub-second latency.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

- **Amazon Campus Prep Series** : Career program centered on technical prep, leadership, and innovation at Amazon.
- **AMD Developer Program** : Completed technical workshops on hardware architecture and compute optimization.
- **PayPal Career Academy** : Participated in fintech engineering prep focused on scalable backend systems.
- **JPMC Data for Good Hackathon** : Collaborated in a team to build scalable data solutions for non-profit challenges.
- **Microsoft Emerging Leaders Program - Asynchronous Cohort**: Leadership and professional storytelling cohort with 20+ peers.
- **AI4ALL Ignite Fellow** : AI-focused innovation program emphasizing responsible AI development and real-world applications.
- **NSBE Director of Corporate Sponsorship**: Professional development, networking, and community for black engineers.
- **CodePath Tech Fellow** : Completed 10-week intensive in network defense, SIEM, and incident response.
- **Intramural Soccer team captain (Grambling State University)** – MVP 2025/26 : Leadership, teamwork, and strategic execution.